Cognitive Transport Protocol (CTP)
Technical Standard Specification (TSS)
This document is frozen as version 1.0 — Reference Standard
Status: Public Reference Specification
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Scope: Standards-Track Specification with Non-Normative Reference Implementation

1. Purpose and Scope

The Cognitive Transport Protocol (CTP) defines a formal control layer for regulating information flow between computational systems and human users.

CTP addresses a structural mismatch in modern human–computer interaction: machine output scales faster than human cognitive recovery capacity. This mismatch produces cognitive congestion, resulting in avoidance, abandonment, degraded decision quality, and systemic risk in high-stakes environments.

This specification formalizes CTP as a Cognitive Regulation Layer ("Layer 8"), extending conventional application-layer architectures with mathematically grounded constraints derived from human processing limits.

This document defines:
Core variables and governing equations
Required protocol mechanisms
Compliance criteria and diagnostic dimensions

This specification does not define:
UI design guidelines
Clinical or diagnostic claims
Proprietary calibration methods
Commercial APIs or datasets

2. Design Principles

CTP is governed by the following principles:

Capacity Is State-Dependent
Human processing capacity varies over time and context and must not be treated as static or infinite.

Avoidance Is a Rational Control Response
Disengagement reflects overload, not user error.

Overload Accumulates Predictably
Cognitive distress integrates over time as a function of interpretative demand and available capacity.

Prevention Outperforms Recovery
Systems must intervene before overload lengthens recovery time.

3. Fundamental Control Variables

CTP defines three primary variables:

3.1 Bandwidth (B)

Definition:
Usable cognitive processing throughput at time t.

Properties:

State-dependent
Distinct from intelligence, skill, or motivation
Reflects effective capacity under current conditions

3.2 Prediction Error Energy (E)
Definition:
Accumulated mismatch between expected and actual cognitive state.
Properties:
Increases when demands exceed available bandwidth
Integrates over time if unmitigated

3.3 Precision (Pi)
Definition:
A control parameter representing the interpretative weight assigned to information.
Properties:
Higher precision increases urgency and cognitive cost
Elevated precision slows recovery
Precision is independent of informational "importance"

4. Governing Equations
4.1 Distress Objective
CTP evaluates system impact using the Distress Objective defined as:

D += Pi * E * dt

(illustrative, non-normative)

5. Control Obligations
5.1 Stabilization Priority
Interfaces must prioritize reduction of uncertainty and system stabilization before presenting high-precision information.
High-precision output during elevated error states constitutes non-compliance.
5.2 Effective Bandwidth Scaling
Systems must scale complexity relative to effective bandwidth:
Failure to do so produces predictable abandonment ("Dashboard Paradox").

6. Diagnostic Framework
CTP compliance is evaluated across four diagnostic dimensions:
Precision Load
Is interpretative demand appropriate to current state?
Bandwidth Assumptions
Does the system assume static or unlimited capacity?
Error Accumulation
Does the system allow unbounded integration of error energy?
Truncation Availability
Are safe exits provided to arrest overload?

7. Compliance and Conformance
This specification defines architectural compliance, not behavioral guarantees.

A system is considered CTP-aligned if it:
Implements the defined variables
Honors the governing constraints
Provides truncation and stabilization mechanisms
Formal certification, calibration thresholds, and empirical validation are outside the scope of this document.

8. Non-Goals and Exclusions

CTP does not:
Diagnose cognitive or psychological conditions
Replace application-level UX design
Define ethical, legal, or regulatory policy
Impose user-specific profiling requirements

9. Versioning and Governance

This document represents CTP v1.0.
Future revisions may extend:
Variable estimation methods
Interoperability guidance
Formal compliance profiles
Backward compatibility and semantic stability are intended design goals.

10. Summary

The Cognitive Transport Protocol formalizes a missing control layer in modern computing systems: the regulation of information flow relative to human recovery limits.
By treating cognitive load as a governable transport problem, CTP enables systems that preserve engagement, safety, and decision quality as machine intelligence scales.